

INTRODUCTION

I am currently a Ph.D candidate at the School of Mathematical Sciences and Institute of Natural Sciences, Shanghai Jiao Tong University, majoring in computational mathematics.

My advisors are Prof. Wenjun Ying and Prof. Weizhu Bao. My research mostly focus on the computational methods and analysis for geometric flows.

EDUCATION

- **SHANGHAI JIAO TONG UNIVERSITY** Shanghai, China
Ph.D in Computational Mathematics
Advisor: Wenjun Ying & Weizhu Bao. Sep 2021 - Jun 2026
- **NATIONAL UNIVERSITY OF SINGAPORE** Singapore
Visiting Ph.D student, Computational Mathematics
Advisor: Wenjun Ying & Weizhu Bao. Oct 2023 - Oct 2025
- **JILIN UNIVERSITY** Jilin, China
B.Sc., Pure and Applied Mathematics Sep 2017 - Jun 2021

RESEARCH INTEREST

- **Geometric PDE:**
Computational methods and analysis for geometric flows, solid-state dewetting, differential equations on manifolds.
- **Multiscale Modeling:**
Multiscale modeling, simulation and applications; phase-field model and interface problems in material sciences.
- **Machine Learning:**
AI4S, data-driven methods.
- **Scientific Computing:**
Numerical analysis, scientific computing and applied mathematics in general.

PUBLICATION LIST

6. **A structure-preserving parametric finite element method for anisotropic surface evolution via an α -surface energy matrix**, in preparation.
5. Pengsong Yin, Wenjun Ying, **Yulin Zhang**, Han Zhou. **A kernel-free boundary integral method for elliptic interface problems on surfaces**, *J. Comput. Phys.*, Vol 562 (2026), article 115004 (arXiv:2508.16061).
4. Weizhu Bao, Yifei Li, Wenjun Ying, **Yulin Zhang**. **A structure-preserving parametric approximation for general anisotropic geometric flows via an α -surface energy matrix**, *J. Comput. Phys.*, Vol 561 (2026), article 114976 (arXiv:2512.24875).
3. Yifei Li, Wenjun Ying, **Yulin Zhang**. **A structure-preserving parametric finite element method with optimal energy stability condition for anisotropic surface diffusion**, *J. Sci. Comput.*, Vol 104, article 76 (arXiv:2501.16660).
2. Fuchang Huo, Weilong Mo, **Yulin Zhang**. **A locking-free conforming discontinuous Galerkin finite element method for linear elasticity problems**, *J. Comput. Appl. Math.*, Vol 465 (2025), article 116582.
1. **Yulin Zhang**, Yifei Li, Wenjun Ying. **A stabilized parametric finite element method for surface diffusion with an arbitrary surface energy**, *J. Comput. Phys.*, Vol 523 (2025), article 113695 (arXiv:2404.02083).

VISITING POSITION

- **National University of Singapore** Singapore
Visiting PhD student Oct 2023 - Oct 2025
 - **Host:** Weizhu Bao.
 - **Topic:** Computational methods and analysis for geometric flows.
- **Jilin University** Changchun, China
Visiting PhD student Mar 2023 - Apr 2023
 - **Host:** Ran Zhang, Qilong Zhai.
 - **Topic:** Conforming discontinuous Galerkin finite element method.

HONORS AND AWARDS

- Outstanding Graduate of Shanghai Jiao Tong University 2026
- National Scholarship for Graduate Students 2025
- SJTU Academic Scholarship for Doctoral Students 2021 - 2026
- SJTU Outstanding Teaching Assistant Award 2022 spring
- SJTU Zhiyuan Honors Doctoral Program 2021 fall - 2026 spring
- JLU Tang Aoqing Honors Program in Maths (Mount Qomolangma Project) 2017 fall - 2021 spring

PROFESSIONAL SERVICE

- **Journal Reviewer:** Journal of Computational Physics

TALKS AND SEMINARS

The 1st National High-Level Doctoral Academic Forum of SJTU	Nov 2025, Shanghai.
The 3rd Doctoral Forum on Frontiers in Mathematics at PKU	Nov 2025, Beijing.
The 14th SIAM Student Chapter Symposium at NUS	Apr 2025, Singapore.
SJTU-NUS PhD Forum for Mathematics	Nov 2024, Singapore.

LANGUAGES

Chinese (native), English.
Python, Matlab, Wolfram Mathematica.